

(1b) $\left[\frac{Fe}{H}\right] = \log\left(\frac{Z}{Z_0}\right)$, Z : object
 Z_0 : Sun $\leftarrow$ Notes

(2) trig parallax: go over Friday

(3) IMF: # stars w mass between $(M, M + \Delta M)$

$$\xi(M) \Delta M \equiv \frac{\text{Vol}}{\text{Vol}}$$

$$\xi(M) = \xi_0 \left(\frac{M}{M_0}\right)^{-\alpha} = \xi_0 \left(\frac{1}{M_0}\right)^{-\alpha} \cdot M^{-\alpha} = \xi_0 M^{-\alpha}$$

$$\alpha \approx 2.35$$

$$\oplus$$

$$\alpha \approx 1.7$$

$$\xi_1(M) = \xi_1 M^{-1.7}$$

$$\xi_2(M) = \xi_2 M^{-2.35}$$

$$[\xi(M) dM] \sim \frac{N_M}{V}, \quad V=1$$

• Polytropic: Equations of state:

$$P \sim \rho^\alpha \quad : \quad dF = \frac{G M dM}{R^2}$$

$$F/a \sim P.$$

* Similar: The gases in different regions have different equations of states:

$\rightarrow$ Related but not nec. the same α .

• Distributions: $\xi(M) dM$

Densities

$$\int \xi(M) dM$$

$$= 1$$

Can find

$$\xi_0$$

by

the

above

$$M_B$$

$$\xi_0$$

$$M^{-\alpha}$$

$$dM$$

$$=$$

$$\xi_0$$

$$\cdot$$

$$\left(\frac{2}{-\alpha+1}\right)$$

$$M^{-\alpha+1}$$

$$\Big|_{M_1}^{M_B}$$

$$= 1$$

$$M_1$$

$$= 1$$

$$M_B$$

$$= 1$$

$$M_1$$

$$= 1$$

When we want to find

$$\xi_0$$

$$[\xi(M)] = \frac{N}{\sqrt{M}} \quad \text{Normalization constant}$$

(3)

$$\xi(M) = \xi_0 M^{-1.7}$$

$$\int_{M_A}^{M_B} \xi(M) dM = N : \text{total \# of stars}$$

$$\xi_0 \int_{M_A}^{M_B} M^{-\alpha} dM = N$$

$$\underbrace{\int_{M_A}^{M_B} M^{-\alpha} dM}_{\tilde{M}} \quad \Bigg| \quad \xi_0 = \frac{N}{\tilde{M}}$$

$$\xi(M) = \frac{N}{\tilde{M}} M^{-\alpha} \quad (\text{only for normalized})$$

Take int.

$$\int \xi(M) dM = 1$$

$$\int_{M_A}^{M_B} \xi(M) dM = N_{\text{tot}}$$

$$\text{Fraction: } \int_{M_1}^{M_2} \xi(M) dM = N_{\text{fract.}}$$

$$\begin{array}{l|l} M_A = 0.5 & M_1 = 3 \\ M_B = 150 & M_2 = 150 \end{array}$$

$$\bullet (\text{Fraction}) = \frac{N_{\text{fract}}}{N_{\text{tot}}} = \frac{\cancel{\xi_0} \int_{M_1}^{M_2} M^{-1.7} dM}{\cancel{\xi_0} \int_{M_A}^{M_B} M^{-1.7} dM}$$

(PS2 Opt a) : solutions

• Want to know average:

$$\text{Exam A} : \alpha = -2.35$$

$$M \geq 30 M_{\odot} \quad (\text{Hypernovae})$$

$$8 < M < 30 \quad (\text{Type II})$$

$$\text{ratio of } \boxed{\frac{\text{Hyper.}}{\text{Type II}}} = \left[\frac{\left(\frac{\text{Hyper}}{\text{Total}} \right)}{\left(\frac{\text{Type II}}{\text{Total}} \right)} \right]$$

$$(\text{Total}) : M : 0.5 - 120 M_{\odot} *$$

$$\text{Type II} : \int_8^{30} \zeta_0 M^{-2.35} dM$$

$$\text{Hyper.} : \int_{30}^{120} \zeta_0 M^{-2.35} dM$$

Recall: $\langle E \rangle = \sum_n P(n) E_n$

$\hookrightarrow$ Cont.

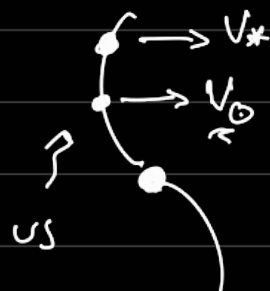
$$\langle f(M) \rangle = \int f(M) \underbrace{\zeta(M)}_{\text{tells us } (M, M + \Delta M)} dM$$

$$\langle M \rangle = \int_{0.5}^{150} M \left(\underbrace{\zeta(M)}_{\text{tells us } (M, M + \Delta M)} dM \right)$$

$$= \int_{0.5}^{150} M \left(\zeta_0 M^{-2.5} \right) dM$$

$$= \zeta_0 \int_{0.5}^{150} M^{-1.5} dM$$

* Milky way: stars we observe all rel. to us



* Looking at the Local Standard Rest (LSR)
the avg. velocities of stars near US.

• S.M. $V_{rms} = \sqrt{\langle V^2 \rangle}$
• Same for stars near US

Nearby old stars: $V_{rms} \sim 20-30 \text{ km/s}$
young $\sim 10 \text{ km/s}$

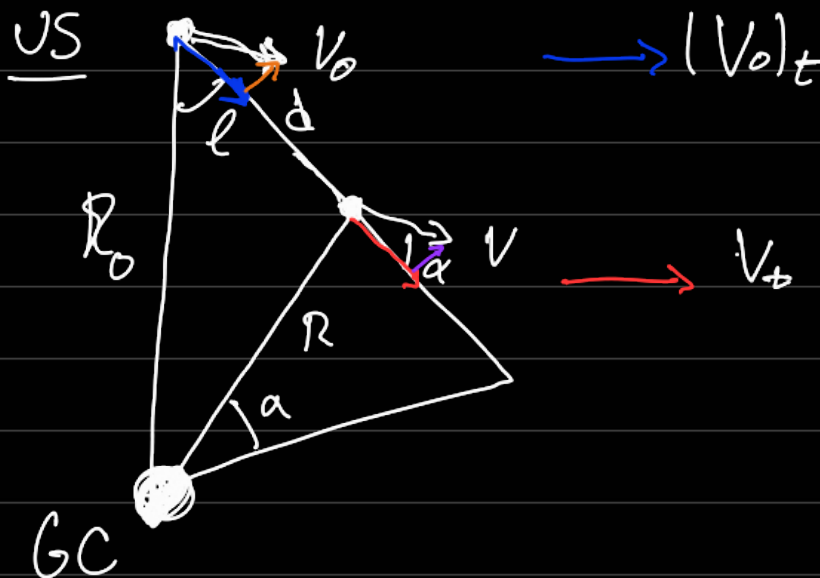
$\sigma^2 \Rightarrow V_{rms} \sqrt{\sigma^2}$

* Different methods for observing these velocities
 $\rightarrow$ Some are empirical

ex:

$$V_t (\text{km s}^{-1}) = 4.74 d(\text{kpc}) \mu (\text{milliarc yr}^{-1})$$

*



$\rightarrow (V_0)_t = \frac{dV_0}{dt}$
* Oort's constants: $\underline{\omega_0},$

P Exam A.2.1

